

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, Java, C++, SQL

Tools/Frameworks: Apache Spark, Apache Flink, TensorFlow, PyTorch, Docker, Kubernetes, FastAPI, Airflow, Databricks, Hopsworks, Triton, SageMaker

Competence: Machine Learning, MLOps, Real-time Inference Systems, Data Engineering, Model Serving, Workflow Automation, Feature Engineering, CI/CD, Distributed Systems, Data Streaming

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, improving model performance and training efficiency.
- Developed and optimized physics-informed hybrid loss functions to enhance model accuracy and stability.
- Collaborated with researchers to publish findings at the FLOMEKO conference.
- Built dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased inference speed by 5x using optimized AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute cost and latency by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Designed and implemented POCs using YOLO, few-shot learning, and CNNs for client demonstrations.
- Conducted educational sessions on machine learning and big data at universities, mentoring students and professionals.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and large language models.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Developed web scrapers and ETL pipelines for structured and unstructured data ingestion.
- Automated data cleaning workflows to improve data quality for client analytics use cases.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Designed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and taxonomic bias mitigation.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system using LSTMs and time-series analysis, achieving high prediction accuracy.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Collaborated to create the world's first AI-driven self-playing Chitralli Sitar, integrating signal processing and music generation models.
- Project featured on Coke Studio in a music video by Hassan Raheem.

Hugging Face Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence *Erlangen, DE*

- Specialized in MLOps, real-time inference systems, and distributed machine learning pipelines.
- Coursework includes Advanced Machine Learning, Data Engineering, and Cloud Computing.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor for academic excellence.
- Capstone project focused on real-time Bitcoin price forecasting using LSTMs and time-series analysis.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python